

Homework #3 (Submit before 20-Apr)

❖ Ch2_No.11

The cavity of a CO₂ laser is formed by a square plane and concave mirror. The cavity length $L=50$ cm, the radius of the curvature of the concave mirror $R=2$ m, the size of the mirror $2a=1$ cm, and the wavelength of the laser $\lambda=10.6$ μm . Calculate w_{s1} , w_{s2} , w_0 , θ_0 , δ_{00}^1 and δ_{00}^2 .

❖ Ch2_No.13

There is a plane mirror and a concave mirror with a curvature $R=1$ m. How to construct a stable cavity which has the minimum far-field divergence angle of the fundamental mode? Draw the curve of relationship between the divergence angle and the cavity length L .